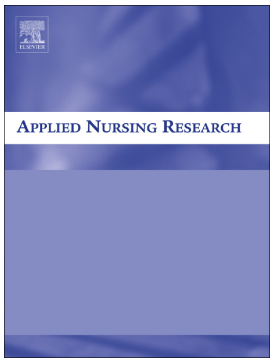
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Development of a Theory-guided Storytelling Narrative Intervention to Improve HPV Vaccination Behavior: Save Our Daughters from Cervical Cancer

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Introduction

HPV vaccination uptakes among ethnic Genital HPV infection is the most common sexually transmitted infection in the U.S., ranging in prevalence from 14% to 90% (Markowitz et al., 2014) and is the cause of a number of genital cancers: cervical cancer, cancers of the vulva, vagina, and penis and cancers of the anus and oropharynx as well as the cause of anogenital warts (Center for Disease Control Prevention [CDC], 2016; Forman et al., 2012; Veins et al., 2016). Studies have confirmed that 99% of cervical cancer cases are linked with HPV (CDC, 2016; Viens et al., 2016) and that Southeast Asian American women have a higher incidence rate of cervical cancer than most other racial and ethnic groups in the U.S. (Kem & Chu, 2007; McCracken et al., 2007). The age-adjusted invasive cervical cancer incidence rate among Cambodian American women (15.0/100,000) was twice the incidence rate among non-Latina white women (7.7/100,000) (Kem & Chu, 2007). Cervical cancer is preventable with regular screening tests and vaccinations and vaccines for HPV have proven effective in preventing HPV-related cancers. (Markowitz et al., 2014; Viens et al., 2016) However, despite an excellent safety and efficacy profile, HPV vaccination rates are low in general and vaccination uptakes among Asian adolescents are lower than for all other groups combined (12.4% vs. 26.8%) (Nomura & Rahman, 2014).

The need to reduce the health disparities related to the prevalence of cervical cancer and low HPV vaccination uptakes among minority women has convinced researchers, clinicians, and community health leaders of the necessity to develop effective health interventions. Both cultural and language-specific HPV-related health interventions as well as rigorous testing of their effectiveness must be developed to tackle the problem of HPV infection and subsequent cervical cancer. Therefore, the purpose of the paper was to develop a novel, theory-guided narrative

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intervention modifying variables influencing human papilloma virus (HPV) vaccination behaviors among Cambodian Americans.

When the vaccine is given to young women less than 18 years of age, parental permission is required. Mothers usually have made decision for her minor daughter to receive the initial HPV vaccine.(Gross & Howard, 2001; Hertweck et al, 2013) Previous studies have shown that parents' knowledge, attitude, and beliefs influence their decision making process regarding health care for their adolescent daughters (Case & Paxon, 2002; Dancy, Crittenden, & Talashek, 2006; Hertweck et al., 2013; Hutchinson et al., 2012). In studies done with White, Black, and Hispanic/Latino adolescent daughters, those whose mothers had more knowledge about their children's health issues and who talked to their children about vaccination and sex, had higher vaccination rates (Dancy, Crittenden, & Talashek, 2006; Hutchinson et al., 2012; McRee, Reiter, Gottlieb, & Brewer, 2011; Murphy, Roberts, & Herbeck, 2011). However, in our previous studies, we found that Cambodian American mothers had a very low level of awareness or knowledge about HPV infection and also faced cultural barriers in communicating with their English-speaking daughters (Lee, Kiang, Kim, et al., 2015; Lee, Kiang, Tang, et al., 2015; Lee et al., 2016). In addition, Cambodian American mothers believed that their English-speaking daughters had better knowledge of health issues than they did. Cambodian American children within immigrant and refugee families typically learn the English language and American culture faster than their parents and their parents have become dependent on their children to communicate with health care providers (Lee, Kiang, Kim, et al., 2015; H. Lee, Kiang, Tang, et al., 2015; Lee et al., 2016). Cambodian American mothers face the challenge of communicating about health issues with their daughters and have little knowledge or experience to draw upon.

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There is clearly a need for health education on HPV infection and HPV vaccination among Cambodian American mothers to increase their daughters' HPV vaccine uptakes. Studies have revealed that narrative/storytelling-based interventions are particularly effective in improving health education with those racial/ ethnic minority populations such as Cambodian Americans that possess a rich tradition of storytelling (Houston et al., 2011; Kreuter et al., 2007; Larkey & Hecht, 2010; Lee, et al, 2015). The advantage of a narrative method over alternative informational approaches includes more effectively reaching an audience who tends to be less involved and has little knowledge about the health issues that may affect them (Downs, 2014; Kreter et al., 2010; Larkey & Gonzalez, 2007). However, rigorous studies specially examining the effectiveness of narrative interventions on vaccination behaviors have not yet been examined among Asian American populations. Therefore, based on preliminary studies, the authors developed a theory-guided narrative health intervention to increase HPV vaccination behavior among Cambodian Americans. The objective of this paper is to provide a detailed description of the process of storytelling narrative intervention development (rather than on testing the intervention) to target Cambodian American mothers with adolescent daughters.

The Process of Theory-Guided Narrative Health Intervention Development

The intervention development process is presented in Table 1 and each stage is discussed in detail below.

Stage 1. Why Narrative Health Intervention?

The Institute of Medicine (2002) reported that many Americans do not understand the health information they receive to make informed decisions and act on them. Health communication can influence knowledge and perceptions that change social norms and motivate action and it can model the desired behavior (National Cancer Institute, 2002). Storytelling, as a

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mode of narrative communication, is deeply rooted in Asian cultures and has been used to transmit ideas in education and history (Lee, Hann, Yang, & Fawcett, 2011; Lee et al., 2014; Lee et al., 2016; Pescosolido, Gardner, & Lubell, 1998). Specifically for the Cambodian people, their literary tradition was mainly oral and most literary works were passed down by monks and spiritual leaders, professional storytellers, or older family members with storytelling serving as a primary means to help people understand and make sense of their health and illness (Bunly, 2005; May, 2015). Cambodian American mothers often have limited English proficiency, poor health literacy, and face cultural barriers which present obstacles to effective health communication (Lee, et al., 2014; Lee, Kiang, Tang, et al., 2015; Schyve, 2007). Thus, it can be reasonably inferred that the narrative approach to health communication rather than the alternative informational approach will be more effective in promoting HPV vaccination behaviors among Cambodian American mothers.

Stage 2. Organizing a Theoretical Framework and Identifying Determinants

We selected two theories: the revised Network Episode Model (rNEM) (Pescosolido, 1991: Lee et al., 2012; 2014; 2016) and the Storytelling Narrative Communication theory (Lee et al., 2016) to guide the narrative intervention development for the following reasons. The NEM has been successfully used to examine health decision-making including vaccination behaviors. (Pescosolido, 1991; Lee et al., 2014; Lee et al., 2016) In the rNEM, an individual's health decision making involves the interplay between sociocultural (social networks and social influences), and individual (belief, knowledge, and attitudes towards HPV infections) factors. The Storytelling Narrative Communication theory (Lee et al., 2016) posits that the cultural relevance and inherent logic of storytelling produces changes in the listener's behavior, attitudes, and motivation through the interaction of transportation, identification, and realism.

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Identification with characters in story increases absorption, reduces counter arguing, and thus enhances the viewer's acceptance of the values and beliefs portrayed in the story. Similarly, according to Bandura's social cognitive theory, individuals learn not only from direct experience but also by observing others and modeling their observed behaviors (Bandura, 2002). Thus, the more viewers/listeners perceive the character in the story and the events involving the character as real, the greater the likelihood they will identify with the characters which consequently increases their engagement with and transportation into the narrative story, and, thereby, changes their HPV vaccination behavior.

In the proposed integrated theoretical approach (figure 1), both individual and socio-cultural-demographic factors were identified as influencing factors or determinants of HPV vaccination behavior. Storytelling narrative communication served as a vehicle of health education intervention which integrates the concepts of "t*ransportation, identification, and realism"* to modify the selected determinants. It was anticipated that improved HPV knowledge and attitudes toward HPV as well as social influences of the mothers in vaccination decision-making would change HPV vaccination uptakes directly or through moderating factors.

Stage 3. Developing Protocol to Create Storytelling Narrative Videos

The narrative intervention was developed collaboratively with Cambodian American community leaders. The storytelling narrative videos, "***Save Our Daughters from Cervical Cancer***" were developed focused on a series of HPV vaccination-related health messages tailored to each of the identified determinants of HPV vaccination within the targeted population of Cambodian American mothers (figure 1). A detailed protocol for creating the storytelling intervention is described below as series of five steps (table 1). The focus of this article is on the development process rather than the effectiveness of our intervention.

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Step 3.1. Selection of "Star" storytellers: Among Cambodian American mothers who participated in our previous qualitative studies (N=15), the research team and Cambodian community leaders selected "star" storytellers who had personal stories reflecting the selected determinants. These "stars" were particularly eloquent, persuasive, and could relate their own experiences in such a way that established a cultural connection with Cambodian American mothers. Four Khmer mothers who have adolescent daughters (both vaccinated and unvaccinated) were selected and invited to be a "star" storyteller. It is important to select storytellers by similarity (Allison et al., 2016; Houston et al, 2011; Fix et al., 2012) since identification with characters in the story increases "homophily" or the perceived similarity between the story's characters and the viewer/audience that enhances their acceptance of the values in the story. Consequently, this may reduce counter-arguing and increase engagement or transportation for the viewer/audience.

Step 3.2. Narrative Collection: The four "star" storytellers were video-recorded in an in-depth, semi-structured individual interview where a variety of stories and experiences were elicited in relation to the determinants of the proposed theoretical framework (figure 1) and the interviews lasted between 30 minutes to one hour. We also asked about the storytellers' experiences (episodes) related to vaccination and whether they were influenced by any suggestion, nagging, or pressure in their decision to vaccinate their daughters. We videotaped real, first person, unscripted narratives to maximize emotional and intellectual engagement. The Cambodian American mothers' stories were told in Khmer and the stories were translated into English, edited, and inserted on screen as subtitles. In addition, we found in our previous studies that Asian immigrant mothers possess not only a low level of health knowledge but also have a misunderstanding of HPV infection and mistakenly think that HPV is a genetic problem, is due

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to unclean hygiene, or that their daughters are too young to receive the vaccination (Lee, Kim, Kiang et al., 2016; Lee, Kiang, Kim et al., 2015). As we expected, some storytellers provided incorrect information about HPV infection and vaccination. We decided to keep these erroneous messages to maintain the overall fidelity and integrity of the storyteller's narrative. However, we provided a means to correct this misinformation for the viewers in the *Learn More* video, which is described below.

Learn More: The Learn More video's purpose was to present clinical information about HPV vaccination by a Khmer (Cambodian) physician who was filmed answering Cambodian American mother's questions. We have learned from our past studies with Cambodian Americans that the processes of interviewing or completing surveys gave the participants opportunities for self-reflection or self-learning. As a result, they expressed their desire to know more about HPV infection and vaccination from credible sources including physicians or health researchers (Lee et al., 2014; Lee, Kiang Kim, et al., 2015; Lee et al., 2016). The findings suggested that viewing storytelling videos might motivate Cambodian Americans to obtain more information and that the Storytelling Narrative Communication theory itself posits that storytelling also allows viewers to feel more emotionally connected to the topic and motivates them to want to learn more about the subject matter. We developed the Learn-More video as a simulated physician-patient communication to provide contextualized information particular to HPV vaccination. Because, although the star storytellers provided "real" stories containing useful and powerful information, they also provided insufficient or wrong information about HPV infection and vaccination. This video not only filled in gaps not covered in the narratives but also counteracted incorrect information provided by the storytellers. The Khmer physician provided evidence-based information that addressed common misunderstandings and cultural

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health beliefs related to cervical cancer and HPV infection within Cambodian American communities and, in addition, used accurate medical terminology both in Khmer and English. Finally, the video strongly emphasized the importance of HPV vaccination and urged Cambodian American mothers to take their daughters to the clinics for HPV vaccinations.

Step 3.3. Narrative Review: The content from the narrative interviews was transcribed and then the scripts of each storyteller were divided into smaller story units. The authors, community health leaders and Khmer research assistants coded the script. Coders rated narrative story units for representation of determinants and HPV vaccination behavior in each story unit. The key story units were then selected based on the integrated theoretical framework in figure 1 and the objectives of HPV vaccination behavior change. Determinants of HBV vaccination and vaccination behavior were addressed by at least one "star" storyteller and the stories were intended to influence at least one determinant of HPV vaccination behavior.

Step 3.4. Narrative Editing: We weaved the most highly rated key story units into a cohesive storyline that portrays the most authentic and relevant (to the research objectives and the Cambodian American mothers' culture) aspects of the storyteller's experience. The first sequence of each storyteller is telling about her cultural background in Khmer to generate emotional connections with viewers. The stories covered the following topics in Storyboards (figure 2): 1) awareness and knowledge of HPV infection, 2) health beliefs; 3) sociocultural context; 4) health communication with daughters and health care providers; 5) testimonies of mothers whose daughters have received HPV vaccinations; and 6) overcoming barriers. We also asked about the storytellers' experiences (episodes) related to vaccination and whether they were influenced by suggestion, nagging, or pressure from their social networks in their decision to get vaccinated. Video sequences of the four star-storytellers were decided based on strength of

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content and emotional engagement according to our integrated theoretical framework and adaptation of previously developed protocols. (Allison et al., 2016).

Step 3.5. Packing the Storytelling Narratives in Video-in DVD: The edited video clips were digitally recorded on DVD-ROM and layered into menus. The four "Star" storytellers' stories were then integrated into one interactive DVD. The screenshot of the main menu: "Save my Daughter" Storyboard is in Figure 3. The DVD contained four "stars" stories and each story was 4-5 minutes in length. The additional video clip of "Learn More" was included. We decided that video on DVDs was the best format to deliver the storytelling narrative intervention because it has been reported that most Asian Americans (including Cambodian Americans) own DVD and spend more time per day watching video DVDs of ethnic movies or drama than watching mainstream TV due to language and cultural barriers (Nielsen, 2016).

Discussion

We have reported a process of the development of a theory-guided intervention using storytelling narrative communication methods to increase HPV vaccination uptake for Cambodian Americans. The development process has included the critical steps of identifying determinants for vaccination behavior and then linking them to culturally relevant narrative health communication intervention to enhance the understanding of HPV infection and vaccination and consequently to change HPV vaccination behaviors. Our narrative interventions emphasized description, explanation, sensitivity to culture, language, and personal experiences. The narratives provided concrete illustrations of determinants which were derived from real stories of Cambodian American mothers and then weaved a sequence of messages within each story, with the narratives of other storytellers, and with the Learn More video. It is our understanding that this is the first theory-guided storytelling narrative intervention to increase

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HPV vaccine uptakes among Asian Americans. Moreover, in light of the effectiveness of the identification by the targeted audience with these culturally grounded experiences, narrative intervention has great potential for changing health behavior in a positive way.

The intervention developed in the current study took a novel approach to enhance Cambodian American mothers' health knowledge and HPV vaccination decisions by providing a culturally and linguistically appropriate storytelling narrative intervention. The Cambodian American community health leaders stated that the images, characters, and messages in the stories were familiar to them and that the health messages were easy to understand. One leader stated, "What she says is relatable to us Khmers in everyday life." And another commented that "it's not artificial [like English TV], it's real, it's our story." These real life stories promoted engagement and/or identification of the listeners/viewers with the star storytellers. This, in turn, facilitated the listener's/viewer's acceptance of the values of HPV vaccination and the suggestions and recommendations of HPV vaccination portrayed in the stories. We have demonstrated the feasibility of the development of culturally-grounded storytelling narrative messages guided by the integrated theory to change HPV vaccine behavior. The next step is to evaluate the effectiveness of the narrative intervention on HPV vaccine uptakes.

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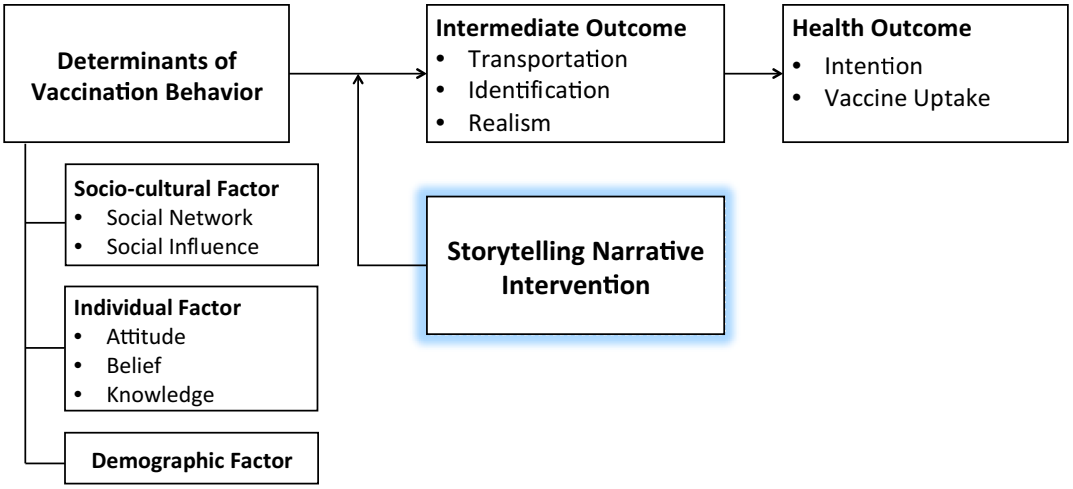
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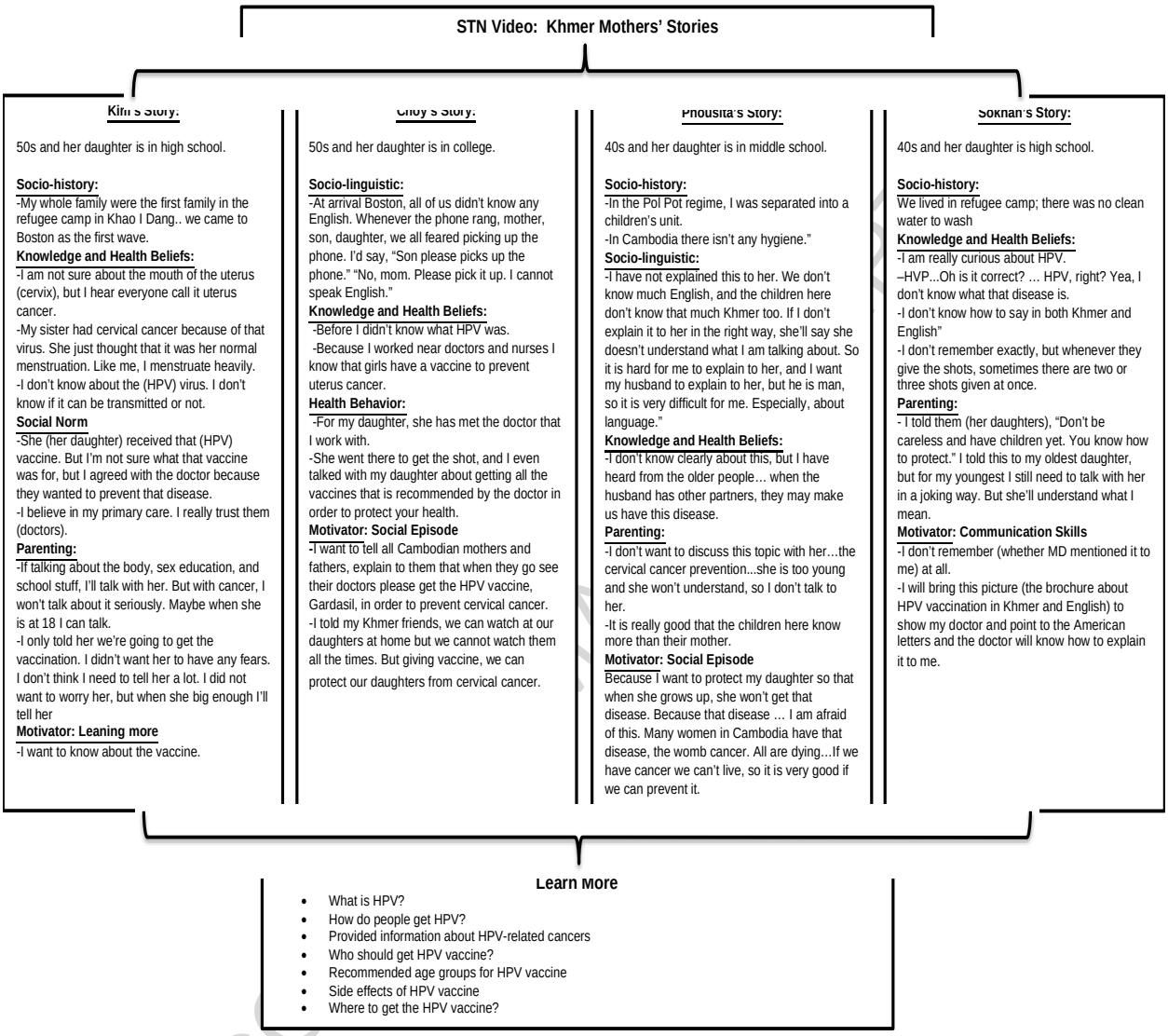
Figure 1. Conceptual Framework to Guide Storytelling Narrative Intervention Development



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Figure 2. Storyboards of Storytelling Narrative Videos: Save Our Daughters from Cervical Cancer



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Figure 3. A ST-video DVD intervention of Khmer mothers



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Table 1. The Process of Theory-Guided Storytelling Narrative Intervention Development

	The Process of Development
Stage 1	Need assessment
Stage 2	Organizing theoretical framework and identifying determinants
Stage 3	Develop protocol to create storytelling narrative intervention
Step 3.1	Selection of “Star” storyteller
Step 3.2	Narrative collection
Step 3.3	Narrative review
Step 3.4	Narrative editing
Step 3.5	Packing

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